

Chapter 9: Human-in-the-Loop (HITL) Pattern and Human-Centric Decision Systems

Part II: Agentic Architecture Patterns

Book: Agentic AI & RAG Systems Handbook

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Chapter Overview

One of the most dangerous assumptions in Agentic AI is:

- | |
|---|
| <ol style="list-style-type: none">1 If the model can reason,2 the model should decide. |
|---|

Enterprise systems reject this assumption.

Production organizations operate under:

- Regulatory requirements
- Legal obligations
- Risk management frameworks
- Compliance controls
- Audit requirements
- Human accountability

As a result, high-quality enterprise AI systems follow a different rule:

Reasoning does not equal authorization.

The Human-in-the-Loop (HITL) pattern introduces human oversight into agent decision making, ensuring that critical actions remain under human control while allowing AI systems to provide speed, analysis, and automation.

HITL is one of the most important patterns in regulated environments including:

- Finance
- Healthcare
- Defense
- Government
- Compliance

- Enterprise Operations
-

Learning Objectives

After completing this chapter you should be able to:

- Understand HITL architectures
 - Design approval workflows
 - Implement risk-based approval systems
 - Integrate humans into LangGraph workflows
 - Use Temporal signals for approvals
 - Design governance-driven agents
 - Implement escalation paths
 - Architect enterprise-grade human oversight systems
 - Explain HITL in Staff and Principal interviews
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Prerequisites

Required:

- Chapter 1: Generative AI Foundations
 - Chapter 2: Agentic AI Fundamentals
 - Chapter 3: Agent Design Principles
 - Chapter 4: ReAct Pattern
 - Chapter 5: Router Pattern
 - Chapter 6: Planner–Executor Pattern
 - Chapter 7: Reflection Pattern
 - Chapter 8: Corrective RAG Pattern
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1. Introduction

Consider an AI agent that recommends:

1 Delete 10,000 customer accounts.

Would you allow:

1 Reason

2	↓
3	Delete

directly?

Most enterprises would answer:

1	No.
---	-----

Instead:

1	Reason
2	↓
3	Recommendation
4	↓
5	Human Review
6	↓
7	Approval
8	↓
9	Execution

This is the core HITL model.

2. Core Concepts

What Is Human-in-the-Loop?

Definition

Human-in-the-Loop is an architectural pattern where human oversight participates in part of an AI workflow before actions are completed.

The human may:

- Approve
 - Reject
 - Modify
 - Escalate
 - Request clarification
-

Core Principle

1	Agent Recommends
2	

3 Human Authorizes

Enterprise Rule

Always separate:

1 Reasoning

from

1 Execution Authority

3. Why HITL Exists

AI systems can:

- Hallucinate
- Misinterpret
- Extrapolate incorrectly
- Act on incomplete information

Even highly capable systems remain probabilistic.

Enterprise organizations require accountability.

Risk Categories

Low Risk

Examples:

1 Summarize document

May not require approval.

Medium Risk

Examples:

1 Open support ticket

May require review.

High Risk

Examples:

1 Modify customer records

Typically requires approval.

Critical Risk

Examples:

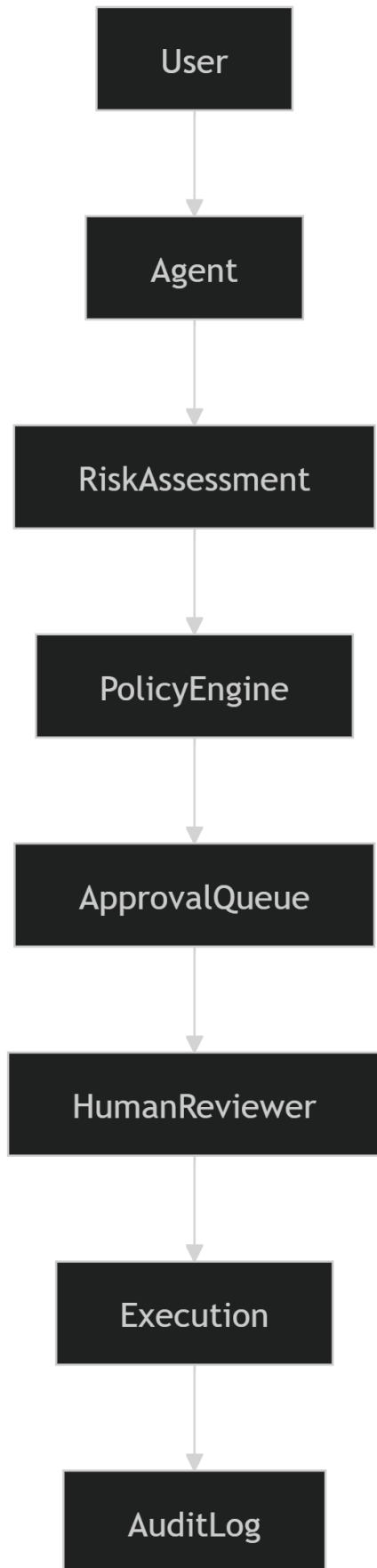
1 Financial transaction
2
3 Data deletion
4
5 Infrastructure changes

Should require mandatory authorization.

4. HITL Architecture

Basic Architecture

Enterprise Architecture



5. Decision Authority Model

Not every decision requires the same level of oversight.

Level 0

Fully Automated

1 Document summarization

Level 1

Human visibility only.

1 Generate reports

Level 2

Human acknowledgment.

1 Customer communications

Level 3

Human approval.

1 Ticket escalation

Level 4

Multi-party approval.

1 Security changes

Level 5

Executive approval.

1 Critical business actions

6. Risk-Based HITL

Risk determines approval requirements.

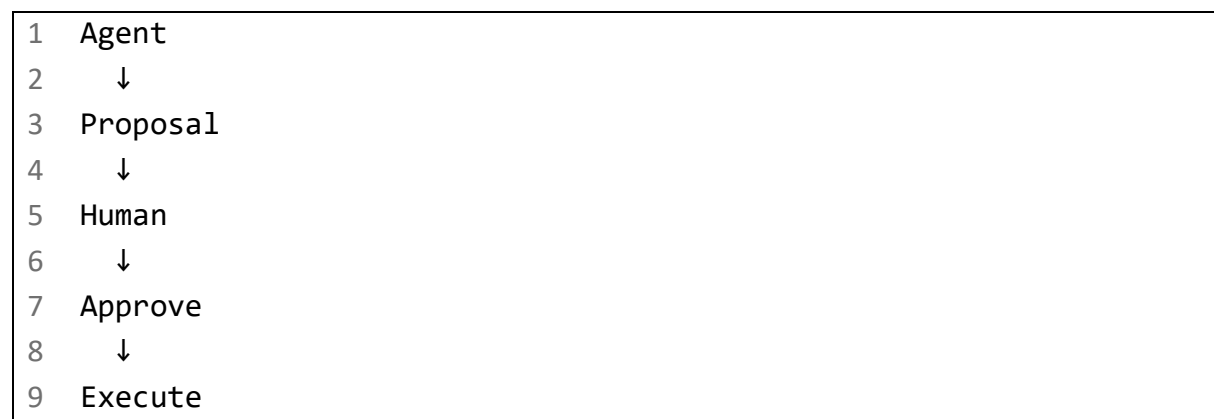
Risk Matrix

Risk Level	Approval Required
Low	No
Medium	Optional
High	Required
Critical	Multi-Level Required

Architecture

7. Approval Workflow Pattern

The most common HITL workflow.



Example

1 Deploy to Production

Agent proposes.

Human approves.

Deployment occurs.

8. Approval Before Workflow Pattern

Instead of approving actions:

Approve workflow execution.

1	Workflow
2	↓
3	Approval
4	↓
5	Execution

Useful for:

- Compliance audits
 - Large migrations
 - Security reviews
-

9. Approval Before Tool Usage

Sometimes tool access itself is risky.

Example:

1	Delete Database
---	-----------------

Tool execution should require:

1	Human Authorization
---	---------------------

before invocation.

10. Escalation Pattern

Certain situations require escalation.

Example:

1	Confidence Low
---	----------------

or

1	Policy Conflict
---	-----------------

Workflow:

Agent



Escalation



Reviewer



Decision

11. Exception Approval Pattern

For unusual situations.

Example:

1 Policy conflict

Agent cannot proceed.

Human determines next steps.

12. Human Approval Interfaces

Approval systems should expose:

Recommendation

1 What is proposed?

Evidence

1 Why?

Confidence

1 How certain?

Impact

1 What changes?

Alternative Actions

1 What other options exist?

13. LangGraph HITL

LangGraph provides first-class support.

Architecture

Interrupt Example

```
1  from langgraph.types import interrupt
2
3  def approval(state):
4
5      approved = interrupt(
6          "Approve deployment?"
7      )
8
9      return {
10         "approved": approved
11     }
```

Benefits

- Durable pauses
 - Human review
 - Safe execution
-

14. Temporal HITL

Temporal excels at long-running human workflows.

Architecture

Temporal Signal Example

```
1  @workflow.signal
2  def approve(self):
3      self.approved = True
```

Benefits

- Days or weeks of waiting
 - Durable state
 - Resume capability
-

15. Multi-Step Approval Workflows

Enterprise approvals often require multiple reviewers.

Example:

```
1 Agent
2 ↓
3 Manager Approval
4 ↓
5 Security Approval
6 ↓
7 Compliance Approval
8 ↓
9 Execution
```

16. SLA-Aware Approvals

Approvals cannot wait forever.

Example:

```
1 approval_timeout:
2   24h
```

After timeout:

```
1 Escalate
```

or

```
1 Reject
```

17. Enterprise Governance Architecture



18. Production Example

Financial Transfer Agent

Goal:

1 Transfer funds

Process:

1 Analyze Request
2 ↓
3 Generate Recommendation
4 ↓
5 Human Approval
6 ↓
7 Transfer Funds

19. DevOps Example

Request:

1 Deploy Production Release

Workflow:

1 Agent Analysis
2 ↓
3 Risk Assessment
4 ↓
5 Approval
6 ↓
7 Deployment

20. Compliance Example

Request:

1 Approve policy exemption

Process:

- | | |
|---|-----------------------------|
| 1 | Review |
| 2 | ↓ |
| 3 | Recommendation |
| 4 | ↓ |
| 5 | Compliance Officer Approval |
| 6 | ↓ |
| 7 | Decision |

21. Security Considerations

Never allow:

- | | |
|---|---------------------|
| 1 | High Risk Action |
| 2 | ↓ |
| 3 | Automatic Execution |

without review.

Controls

Identity Verification

Who approved?

Authorization

Could they approve?

Audit Logging

Was approval recorded?

Tamper Protection

Can approval history be trusted?

22. Observability

Track:

Approval Rate

- | | |
|---|------------------|
| 1 | Approved / Total |
|---|------------------|

Rejection Rate

1 Rejected / Total

Escalation Rate

1 Escalated / Total

Approval Latency

1 Time To Decision

Override Frequency

1 Human Changed Recommendation

23. Reliability

HITL improves:

- Safety
- Governance
- Compliance

But introduces:

- Human delays
- Operational complexity
- Process bottlenecks

Design appropriately.

24. Common Mistakes

Mistake 1

Human approval for every action.

Mistake 2

No approval for critical actions.

Mistake 3

Missing evidence.

Mistake 4

No escalation path.

Mistake 5

No timeout handling.

25. Anti-Patterns

Approval Theatre

Human clicks approve without context.

No value.

Fully Autonomous Critical Actions

High risk.

Hidden Approvals

Lack transparency.

No Audit Trail

Fails governance.

26. Trade-Offs

Benefit	Cost
Higher safety	More latency
Better governance	More complexity
Better compliance	More operational effort
Reduced risk	Slower execution

27. Architecture Decision Matrix

Scenario	HITL Fit
Financial Systems	Essential
Compliance Systems	Essential
Healthcare	Essential
DevOps Production Changes	Strongly Recommended
Enterprise Search	Limited
FAQ Bot	Poor
Summarization	Poor

28. Real-World Use Cases

Financial Operations

Payment authorization.

Healthcare

Treatment recommendation review.

Compliance

Policy exemptions.

DevOps

Production deployments.

Security Operations

High-risk remediation.

29. Interview Questions

Q1

What is HITL?

Q2

Why is HITL important?

Q3

What does “reasoning does not equal authorization” mean?

Q4

What actions require HITL?

Q5

How does LangGraph support HITL?

Q6

How does Temporal support HITL?

Q7

What are common HITL failures?

Q8

How would you design enterprise approvals?

30. Interview Answers

Q1

A pattern where humans participate in AI decision workflows prior to execution.

Q2

To improve safety, accountability, compliance, and governance.

Q3

An agent may reach a conclusion but should not automatically have permission to execute actions.

Q4

High-risk actions such as deployments, financial transactions, and destructive operations.

Q5

Through interrupts and resumable workflows.

Q6

Through signals, timers, durable workflow state, and long-running execution.

Q7

Approval bottlenecks, poor evidence, lack of escalation, and audit failures.

Q8

Risk-based approval architecture with identity, authorization, audit trails, and escalation workflows.

31. Architecture Review Checklist

HITL Readiness

- Risk matrix defined
 - Approval workflow documented
 - Escalation process implemented
 - Timeout policy defined
 - Audit logging enabled
 - Identity verification implemented
 - Authorization controls implemented
 - Observability configured
 - Governance reviewed
-

32. Practical Exercises

1. Design a HITL architecture for financial transfers.
2. Implement LangGraph interrupt-based approvals.
3. Create a Temporal approval workflow.
4. Design a multi-stage compliance review process.
5. Build an approval audit model.

33. Key Takeaways

- HITL is a foundational enterprise AI pattern.
 - Reasoning and authorization must remain separate.
 - Risk should determine human involvement.
 - LangGraph provides interrupt-driven approvals.
 - Temporal provides durable approval workflows.
 - High-risk actions require oversight.
 - Auditability is mandatory.
 - Escalation and timeout handling are essential.
 - Not every workflow needs human approval.
 - Effective enterprise AI balances automation with accountability.
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34. References

- [LangGraph Documentation](#)
 - [Temporal Documentation](#)
 - [Responsible AI Governance Frameworks](#)
 - [OpenAI Safety Guidelines](#)
 - [Anthropic Constitutional AI Research](#)
 - [Enterprise Risk Management Frameworks](#)
 - [NIST AI Risk Management Framework](#)
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Chapter Status: Completed

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Next Chapter: Chapter 10 – Tool Calling Pattern, Tool Governance, and Action Execution Architecture